Exploring Gene Expression in Breast Cancer Samples from the TCGA-BRCA Cohort

Jewel Irvin

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1 Data Preparation

```
# Set the working directory to the Downloads folder
setwd("~/Downloads")

# Check the working directory
getwd()
```

```
## [1] "/Users/jewels/Downloads"
```

```
# Check the files available in the Downloads folder
#list.files()

# Load the data
counts <- read.csv("counts (1).csv", check.names = FALSE)

names(counts)[1] <- "gene_id"
meta_data <- read.csv("meta_data (1).csv", check.names = FALSE)

# Inspect the data
dim(counts)
```

```
## [1] 60660  1232
```

```
dim(meta_data)
```

```
## [1] 1231   98
```

```
names(counts)[1:5]
```

```
## [1] "gene_id"                      "TCGA-GM-A2DL-01A-11R-A18M-07"
## [3] "TCGA-AC-A2QI-01A-12R-A19W-07" "TCGA-A8-A06R-01A-11R-A00Z-07"
## [5] "TCGA-EW-A1PD-01A-11R-A144-07"
```

```r
names(meta_data)
```

```
##  [1] "barcode"
##  [2] "patient"
##  [3] "sample"
##  [4] "shortLetterCode"
##  [5] "definition"
##  [6] "sample_submitter_id"
##  [7] "sample_type_id"
##  [8] "tumor_descriptor"
##  [9] "sample_id"
## [10] "sample_type"
## [11] "composition"
## [12] "days_to_collection"
## [13] "state"
## [14] "initial_weight"
## [15] "preservation_method"
## [16] "pathology_report_uuid"
## [17] "submitter_id"
## [18] "oct_embedded"
## [19] "specimen_type"
## [20] "is_ffpe"
## [21] "tissue_type"
## [22] "synchronous_malignancy"
## [23] "ajcc_pathologic_stage"
## [24] "days_to_diagnosis"
## [25] "laterality"
## [26] "treatments"
## [27] "tissue_or_organ_of_origin"
## [28] "age_at_diagnosis"
## [29] "primary_diagnosis"
## [30] "metastasis_at_diagnosis"
## [31] "prior_malignancy"
## [32] "year_of_diagnosis"
## [33] "prior_treatment"
## [34] "diagnosis_is_primary_disease"
## [35] "method_of_diagnosis"
## [36] "ajcc_staging_system_edition"
## [37] "ajcc_pathologic_t"
## [38] "morphology"
## [39] "ajcc_pathologic_n"
## [40] "ajcc_pathologic_m"
## [41] "classification_of_tumor"
## [42] "diagnosis_id"
## [43] "icd_10_code"
## [44] "site_of_resection_or_biopsy"
## [45] "sites_of_involvement"
## [46] "days_to_last_follow_up"
## [47] "follow_ups_disease_response"
## [48] "race"
## [49] "gender"
## [50] "ethnicity"
## [51] "vital_status"
```

```
## [52] "age_at_index"
## [53] "days_to_birth"
## [54] "demographic_id"
## [55] "age_is_obfuscated"
## [56] "country_of_residence_at_enrollment"
## [57] "bcr_patient_barcode"
## [58] "primary_site"
## [59] "project_id"
## [60] "disease_type"
## [61] "name"
## [62] "releasable"
## [63] "released"
## [64] "figo_stage"
## [65] "figo_staging_edition_year"
## [66] "days_to_death"
## [67] "tumor_of_origin"
## [68] "days_to_sample_procurement"
## [69] "last_known_disease_status"
## [70] "tumor_grade"
## [71] "progression_or_recurrence"
## [72] "alcohol_history"
## [73] "exposure_id"
## [74] "year_of_birth"
## [75] "paper_patient"
## [76] "paper_Tumor.Type"
## [77] "paper_Included_in_previous_marker_papers"
## [78] "paper_vital_status"
## [79] "paper_days_to_birth"
## [80] "paper_days_to_death"
## [81] "paper_days_to_last_followup"
## [82] "paper_age_at_initial_pathologic_diagnosis"
## [83] "paper_pathologic_stage"
## [84] "paper_Tumor_Grade"
## [85] "paper_BRCA_Pathology"
## [86] "paper_BRCA_Subtype_PAM50"
## [87] "paper_MSI_status"
## [88] "paper_HPV_Status"
## [89] "paper_tobacco_smoking_history"
## [90] "paper_CNV.Clusters"
## [91] "paper_Mutation.Clusters"
## [92] "paper_DNA.Methylation.Clusters"
## [93] "paper_mRNA.Clusters"
## [94] "paper_miRNA.Clusters"
## [95] "paper_lncRNA.Clusters"
## [96] "paper_Protein.Clusters"
## [97] "paper_PARADIGM.Clusters"
## [98] "paper_Pan.Gyn.Clusters"
```

```
#head(counts[, 1:5])
#head(meta_data)
```

2)Gene Selection and Summary Statistics

```
## Inspecting the Data

## Selected Genes: ESR1 and ERBB2

#ESR1 was selected as the primary gene for this analysis because it is an important gene involved in es

# Give the first column a consistent name
names(counts)[1] <- "gene_id"

# Find ESR1 by its Ensembl ID
esr1_row <- counts[
  counts$gene_id == "ENSG00000091831.24",
]

# Check the result
#nrow(esr1_row)

#esr1_counts <- as.numeric(esr1_row[1, -1])

# Check the first few values
#head(esr1_counts)

# Identify the ESR1 and ERBB2 rows
esr1_row <- counts[
  counts$gene_id == "ENSG00000091831.24", ]

erbb2_row <- counts[
  counts$gene_id == "ENSG00000141736.14", ]

# Extract expression counts across all samples
esr1_counts <- as.numeric(esr1_row[1, -1])
erbb2_counts <- as.numeric(erbb2_row[1, -1])

# Calculate summary statistics for each gene
summary_stats <- data.frame(
  Gene = c("ESR1", "ERBB2"),
  Mean = c(mean(esr1_counts, na.rm = TRUE),
           mean(erbb2_counts, na.rm = TRUE)),
  Median = c(median(esr1_counts, na.rm = TRUE),
             median(erbb2_counts, na.rm = TRUE)),
  SD = c(sd(esr1_counts, na.rm = TRUE),
         sd(erbb2_counts, na.rm = TRUE)),
  Minimum = c(min(esr1_counts, na.rm = TRUE),
              min(erbb2_counts, na.rm = TRUE)),
  Maximum = c(max(esr1_counts, na.rm = TRUE),
              max(erbb2_counts, na.rm = TRUE))
)

# Round values for presentation
summary_stats[-1] <- lapply(
  summary_stats[-1],
```

```
  round,
  digits = 2
)

# Display the table
knitr::kable(
  summary_stats,
  caption = "Summary Statistics for ESR1 and ERBB2 Gene Expression"
)
```

Table 1: Summary Statistics for ESR1 and ERBB2 Gene Expression

Gene	Mean	Median	SD	Minimum	Maximum
ESR1	31442.22	17561	38261.25	14	270977
ERBB2	46982.49	19125	105287.55	205	995067

```
table_grob <- tableGrob(
  summary_stats,
  rows = NULL,
  theme = ttheme_minimal(base_size = 14)
)

png(
  filename = "~/Downloads/summary_statistics.png",
  width = 1800,
  height = 700,
  res = 200
)

grid.draw(table_grob)
dev.off()
```

```
## pdf
##   2
```

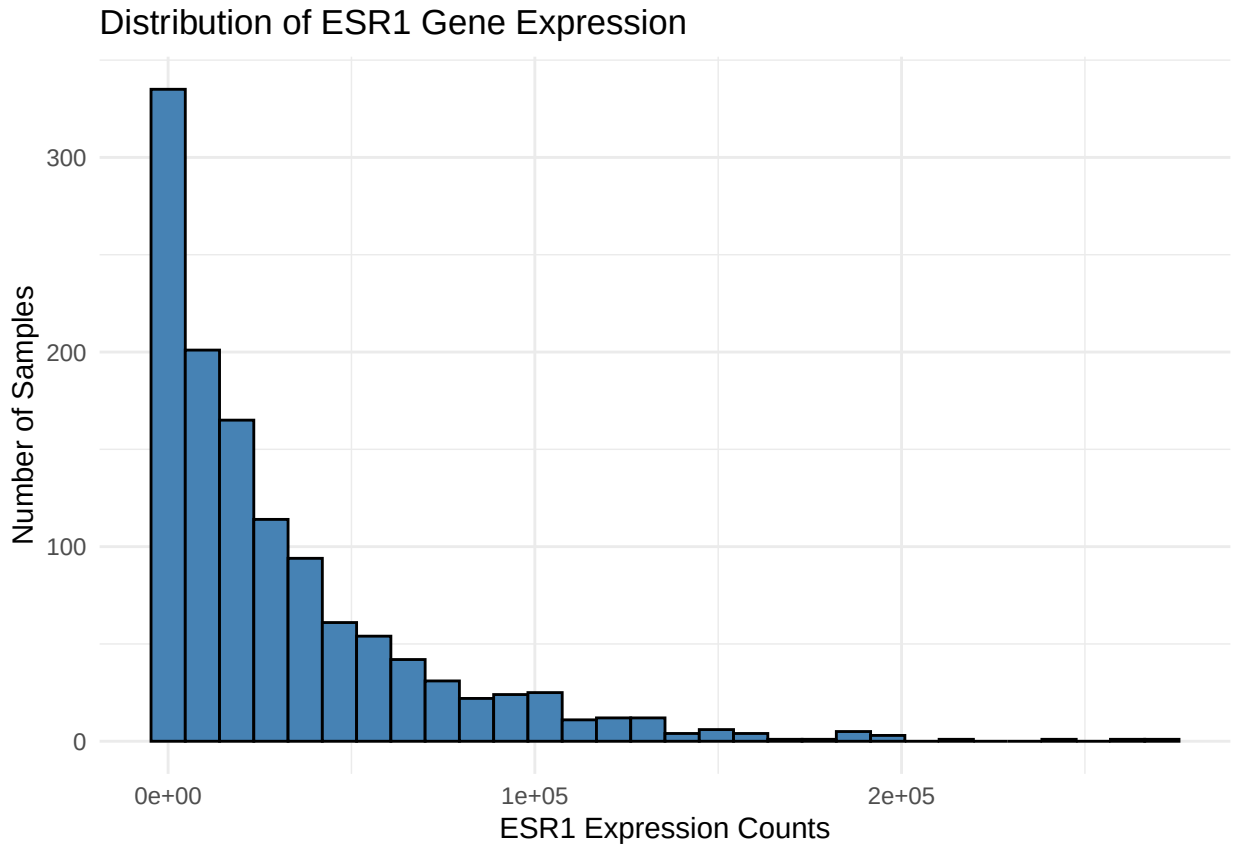
Visualizations

#A) ESR1 Expression Histogram

```
# Create a data frame for ESR1 expression
esr1_df <- data.frame(ESR1 = esr1_counts)

# Create the updated histogram
ggplot(esr1_df, aes(x = ESR1)) +
  geom_histogram(bins = 30, color = "black", fill = "steelblue") +
  labs(
    title = "Distribution of ESR1 Gene Expression",
    x = "ESR1 Expression Counts",
    y = "Number of Samples"
```

```
  ) +
  theme_minimal()
```




```
ggsave("esr1_histogram.png",
       width = 8, height = 5, dpi = 300)
```

#B) Scatter plot — ESR1 vs. ERBB2

```
# Check the actual name of the first column
names(counts)[1]
```

```
## [1] "gene_id"
```

```
# Look at the first few gene IDs
head(counts[[1]])
```

```
## [1] "ENSG00000000003.15" "ENSG00000000005.6"  "ENSG00000000419.13"
## [4] "ENSG00000000457.14" "ENSG00000000460.17" "ENSG00000000938.13"
```

```
# Extract the ERBB2 gene row
erbb2_row <- counts[
  counts[[1]] == "ENSG00000141736.14",
]
```

```r
# Confirm that one row was found
nrow(erbb2_row)
```

```
## [1] 1
```

```r
# Extract expression counts across samples
erbb2_counts <- as.numeric(erbb2_row[1, -1])

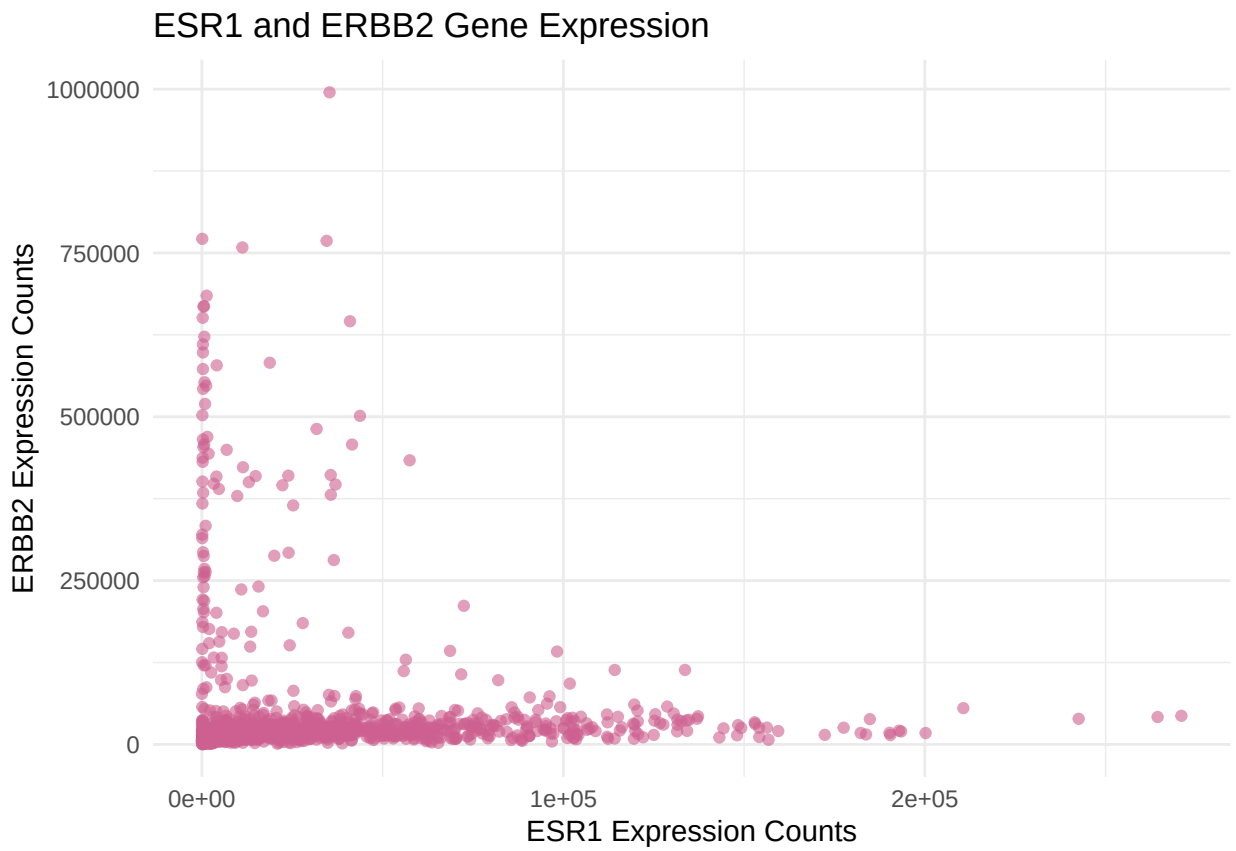
# View the first six counts
head(erbb2_counts)
```

```
## [1]   7613  41006   6858  52772 453696  24483
```

```r
## B) ESR1 and ERBB2 Scatter Plot

# Create a data frame for the scatter plot
esr1_erbb2_df <- data.frame(
  ESR1 = esr1_counts,
  ERBB2 = erbb2_counts
)

# Plot ESR1 versus ERBB2 expression
ggplot(esr1_erbb2_df, aes(x = ESR1, y = ERBB2)) +
  geom_point(color = "hotpink3",alpha = 0.6) +
  labs(
    title = "ESR1 and ERBB2 Gene Expression",
    x = "ESR1 Expression Counts",
    y = "ERBB2 Expression Counts"
  ) +
  theme_minimal()
```




```
ggsave("esr1_erbb2_scatter.png",
       width = 8, height = 5, dpi = 300)
```

#C) Box Plot

```
library(ggplot2)
library(dplyr)

# Extract ESR1 gene expression
esr1_row <- counts[counts$gene_id == "ENSG00000091831.24", ]

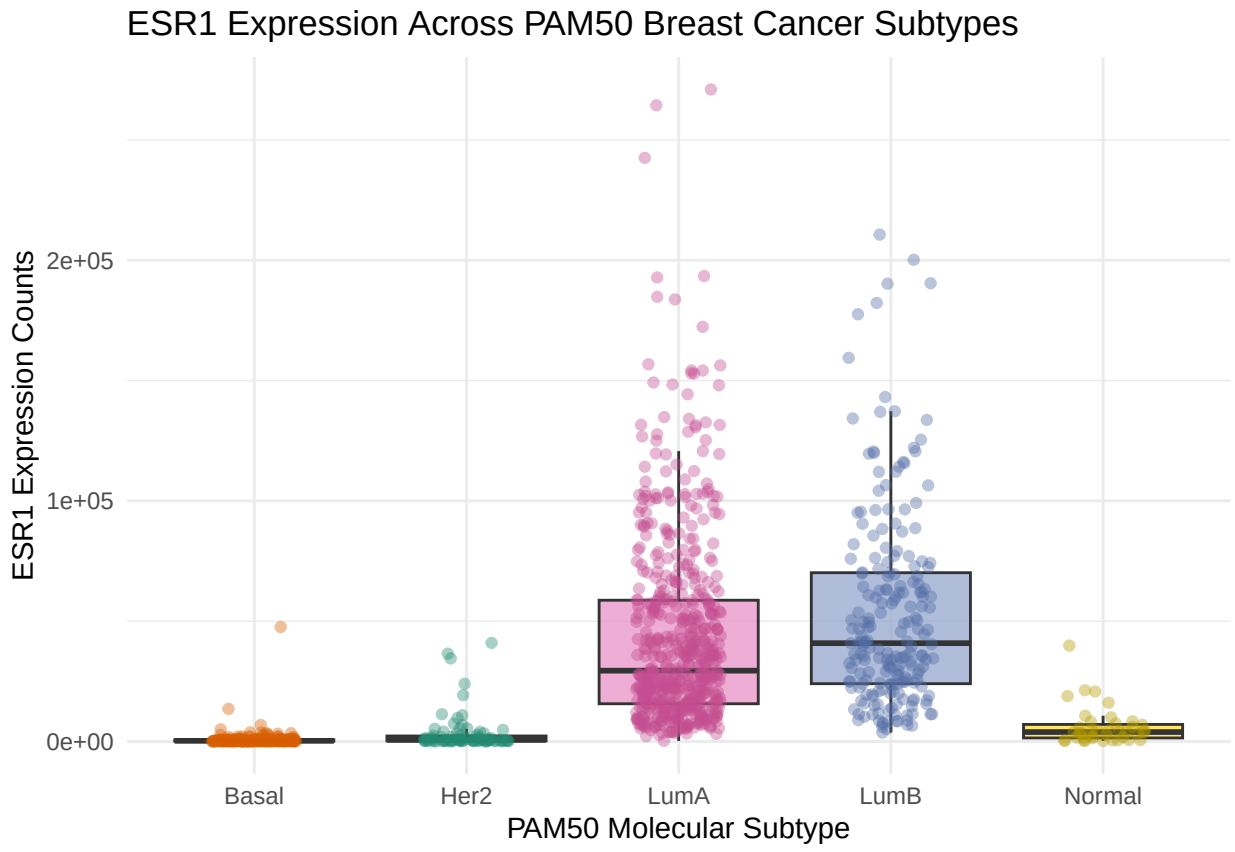
esr1_counts <- as.numeric(esr1_row[1, -1])

# Combine expression data with subtype information
esr1_subtype_df <- data.frame(
  ESR1 = esr1_counts,
  PAM50 = meta_data$paper_BRCA_Subtype_PAM50
)

esr1_subtype_df <- esr1_subtype_df %>%
  filter(!is.na(PAM50))

# Create box plot
ggplot(esr1_subtype_df, aes(x = PAM50, y = ESR1, fill = PAM50)) +
  geom_boxplot(alpha = 0.7, outlier.shape = NA) +
```

```
  geom_jitter(aes(color = PAM50), width = 0.2, alpha = 0.4, size = 1.5) +
  scale_fill_manual(values = c(
    "LumA" = "#E78AC3",
    "LumB" = "#8DA0CB",
    "Basal" = "#FC8D62",
    "Her2" = "#66C2A5",
    "Normal" = "#FFD92F"
  )) +
  scale_color_manual(values = c(
    "LumA" = "#C44E91",
    "LumB" = "#536DA6",
    "Basal" = "#D95F02",
    "Her2" = "#238B72",
    "Normal" = "#B39B00"
  )) +
  labs(
    title = "ESR1 Expression Across PAM50 Breast Cancer Subtypes",
    x = "PAM50 Molecular Subtype",
    y = "ESR1 Expression Counts"
  ) +
  theme_minimal() +
  theme(legend.position = "none")
```




```
ggsave("esr1_pam50_boxplot.png",
       width = 8, height = 5, dpi = 300)
```

#4)Heatmap Analysis

```r
# Load the heatmap package
library(pheatmap)
```

```
##
## Attaching package: 'pheatmap'
```

```
## The following object is masked from 'package:ComplexHeatmap':
##
##     pheatmap
```

```r
# Create the gene expression matrix from the count data
gene_matrix <- as.matrix(counts[, -1])
rownames(gene_matrix) <- counts$gene_id

# Select the 10 most variable genes
gene_variances <- apply(gene_matrix, 1, var, na.rm = TRUE)
top_genes <- names(sort(gene_variances, decreasing = TRUE))[1:10]

# Get the sample IDs from the gene expression matrix
sample_ids <- colnames(gene_matrix)

# Match sample IDs to metadata barcodes
metadata_rows <- match(sample_ids, meta_data$barcode)

# Keep samples with matching metadata and known PAM50 subtypes
keep <- !is.na(metadata_rows)

keep[keep] <- !is.na(
  meta_data$paper_BRCA_Subtype_PAM50[metadata_rows[keep]]
)

# Select the top genes and samples with known subtypes
heatmap_matrix <- log2(
  gene_matrix[top_genes, keep, drop = FALSE] + 1
)

# Create the PAM50 annotation data frame
annotation_col <- data.frame(
  PAM50 = meta_data$paper_BRCA_Subtype_PAM50[metadata_rows[keep]]
)

# Align annotation rows with the heatmap columns
rownames(annotation_col) <- colnames(heatmap_matrix)

# Draw the heatmap
pheatmap(
  heatmap_matrix,
  annotation_col = annotation_col,
  show_colnames = FALSE,
  main = "Gene Expression Across PAM50 Subtypes"
)
```




```
ggsave("heatmap.png",
       width = 8, height = 5, dpi = 300)
```

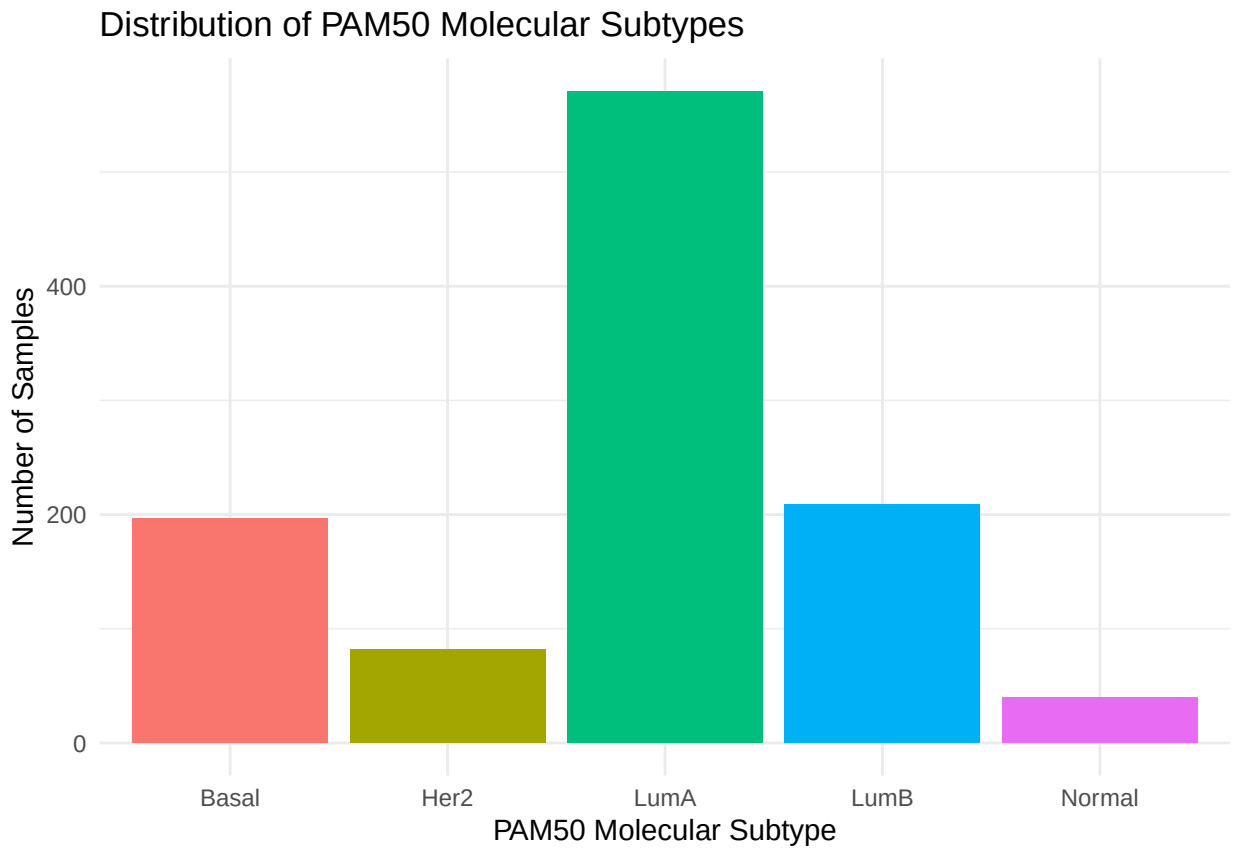
Note that the `echo = FALSE` parameter was added to the code chunk to prevent printing of the R code that generated the plot. #5) Bar Chart

```
library(ggplot2)
library(dplyr)

pam50_counts <- meta_data %>%
  filter(!is.na(paper_BRCA_Subtype_PAM50)) %>%
  count(paper_BRCA_Subtype_PAM50)

ggplot(
  pam50_counts,
  aes(x = paper_BRCA_Subtype_PAM50,
      y = n,
      fill = paper_BRCA_Subtype_PAM50)
) +
  geom_col() +
  labs(
    title = "Distribution of PAM50 Molecular Subtypes",
    x = "PAM50 Molecular Subtype",
    y = "Number of Samples"
  ) +
```

```
  theme_minimal() +
  theme(legend.position = "none")
```




```
ggsave("pam50_subtype_barplot.png",
       width = 8, height = 5, dpi = 300)
```